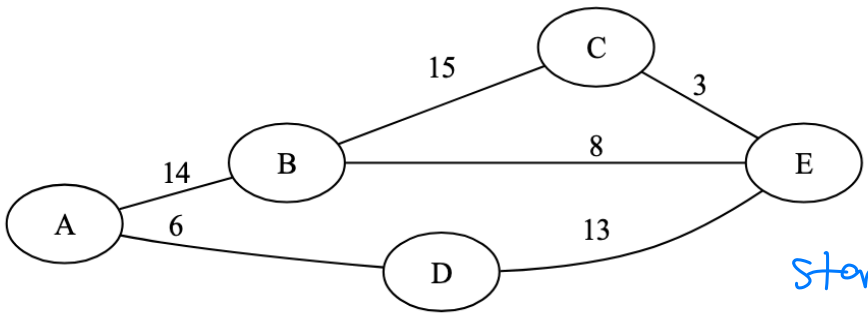
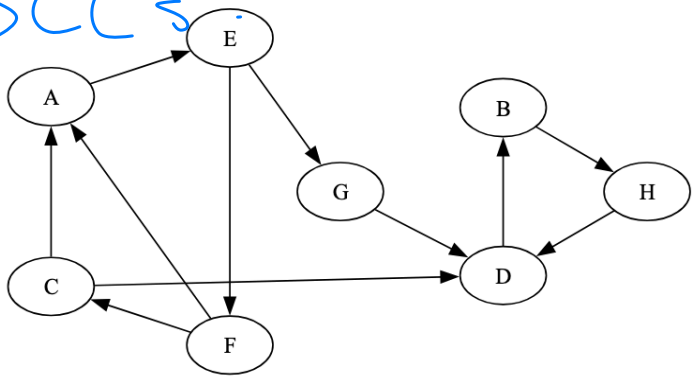
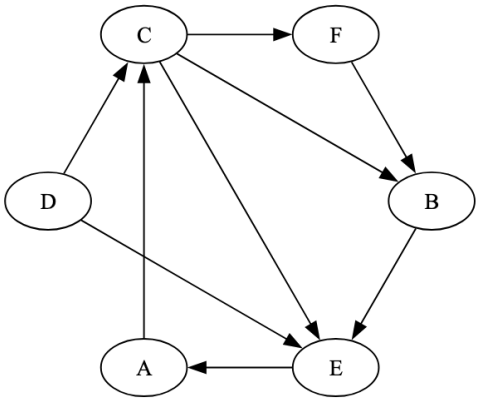
BFS w/ priority queue: total weight along path



start from C

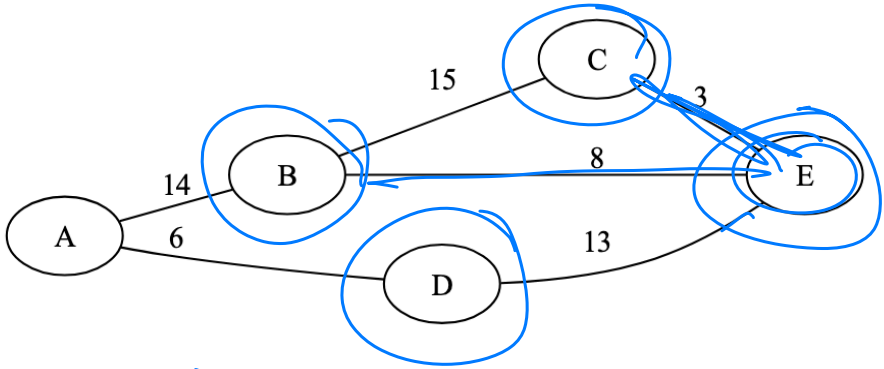
SCCs?





Pre order attributes?

PQ: total sum of weights up to that node



WFS(C) ←

pq:

D - 16

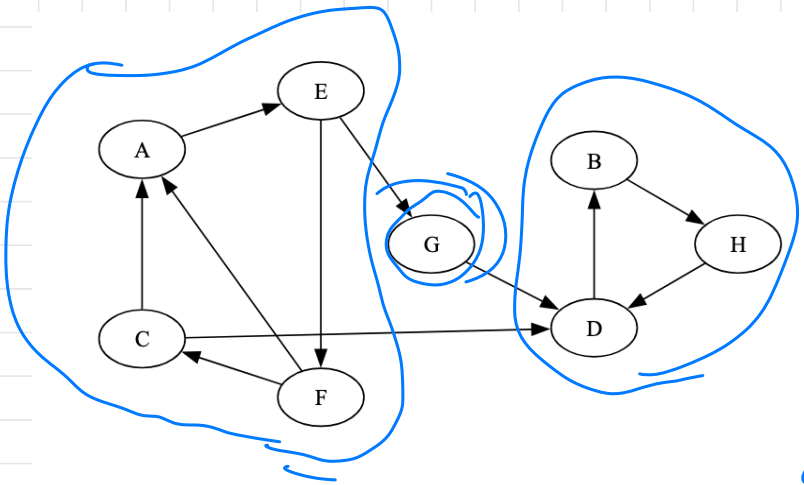
C - 26

A - 25

E - 19

```
WhateverFirstSearch(s):
put s into the bag
while the bag is not empty:
    take v from the bag
    if v is unmarked:
        mark v
        for each neighbor w of v:
            put w into the bag
```

What are the SCCs?



a set of nodes where all are pairwise reachable

are F, H in a SCC?

does F reach H? yes

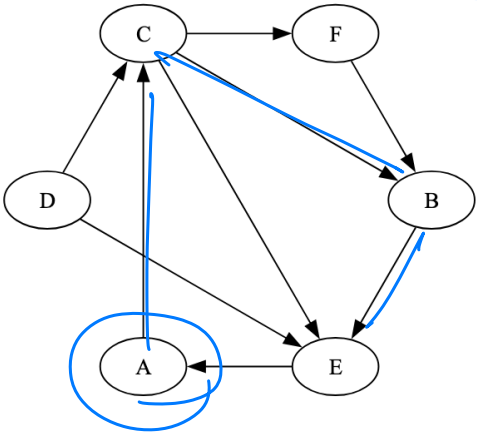
does H reach F? no

{B, D, H}

{A, C, F, E}

{G}

clock - increment anytime a DFS call starts or finishes



n nodes

clock starts at 0

DFSALL(G)

clock = 0

DFS(A)

A.pre = 1

C.pre = 2

B.pre = 3

E.pre = 4

E.post = 5